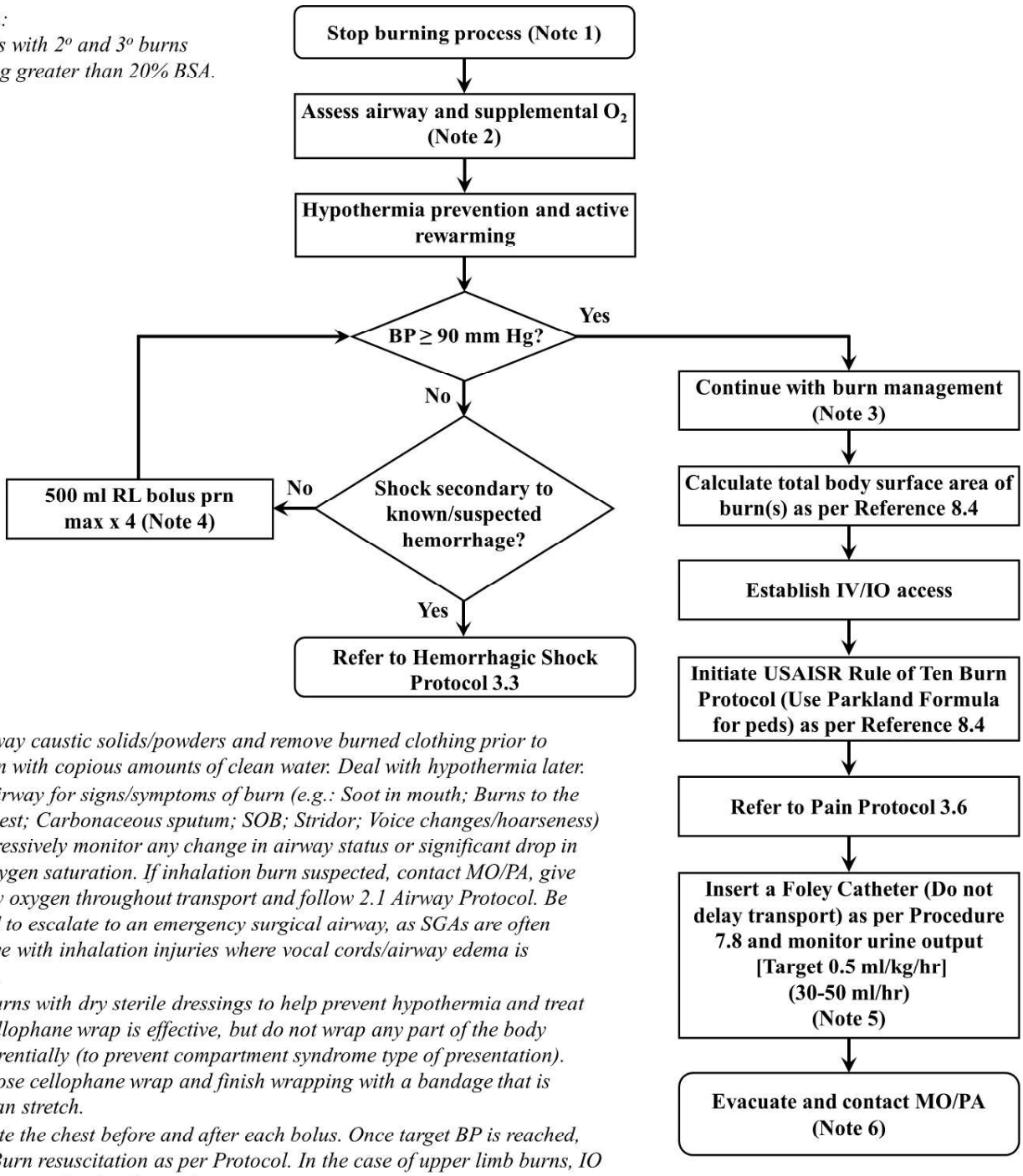
3.5 Burn Management – Class A

Indication:
- *Patients with 2º and 3º burns covering greater than 20% BSA.*



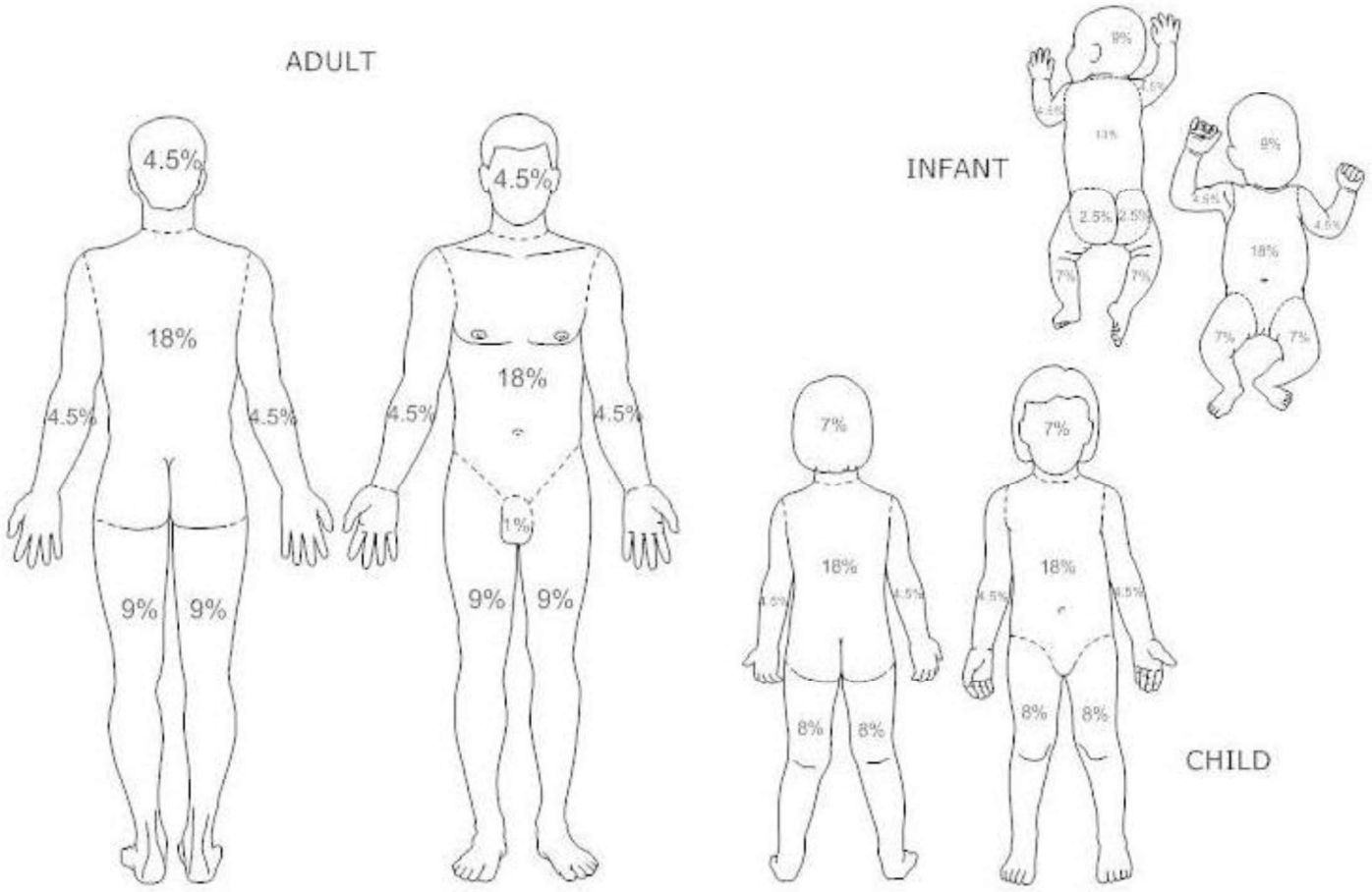
Notes:

1. *Brush away caustic solids/powders and remove burned clothing prior to irrigation with copious amounts of clean water. Deal with hypothermia later.*
2. *Assess airway for signs/symptoms of burn (e.g.: Soot in mouth; Burns to the upper chest; Carbonaceous sputum; SOB; Stridor; Voice changes/hoarseness) and aggressively monitor any change in airway status or significant drop in blood oxygen saturation. If inhalation burn suspected, contact MO/PA, give high flow oxygen throughout transport and follow 2.1 Airway Protocol. Be prepared to escalate to an emergency surgical airway, as SGAs are often ineffective with inhalation injuries where vocal cords/airway edema is common.*
3. *Cover burns with dry sterile dressings to help prevent hypothermia and treat pain. Cellophane wrap is effective, but do not wrap any part of the body circumferentially (to prevent compartment syndrome type of presentation). Apply loose cellophane wrap and finish wrapping with a bandage that is elastic/can stretch.*
4. *Auscultate the chest before and after each bolus. Once target BP is reached, initiate Burn resuscitation as per Protocol. In the case of upper limb burns, IO access may need to be obtained.*
5. *For Prolonged Field Care, fluid resuscitation should be titrated to maintain urinary output (30-50 ml/hr or 0.5 ml/kg/hr). If UOP < 30 ml/hr, increase volume by 25% for the next hour and reassess. If UOP > 50 ml/hr, decrease IV fluid by 25% for the next hour and reassess.*
6. *Antibiotics are not normally used as a prophylactic treatment in the absence of open wounds. If cellulitis develops after several days, contact MO/PA.*

SECTION 8: REFERENCES/ABBREVIATIONS

8.4 Burn Assessment and Fluid Replacement Principles

Rule of Nines Body Surface Area (BSA) Estimation



Fluid Replacement Requirements for Burn Victims

Basic Principles:

- If burns are greater than 20% of Total Body Surface Area (TBSA), fluid resuscitation should be initiated as soon as IV/IO access is established.
- Resuscitation should be initiated with Lactated Ringer's (LR) preferably.
- If Hemorrhagic Shock is present/suspected, resuscitation for Hemorrhagic Shock takes precedence over resuscitation for burns. If in Hypovolemic Shock not associated with suspected Hemorrhage, give IV bolus of 500 ml LR (or Normal Saline if LR unavailable) up to 4 times until BP ≥ 90 mm Hg, then, start the burn fluid resuscitation as per Parkland Formula (Paediatric) or USAISR Rule of Ten (Adult).
- Both calculations (Parkland/USAISR) provide initial IV fluid rate of administration. Adjust fluid rate hourly based on urinary output in order to titrate IV administration (Refer to Prolonged Field Care section below for target urinary output and titration).
- If no IV solution available, use following improvised electrolyte solution to administer orally for patient with a TBSA under 30% (Use same administration/volume rate as IV administration):
    - Oral Rehydration Solution (as per package instructions).
    - Or 1 liter of potable water with 6 level teaspoons sugar and 0.5 level teaspoon salt.

8.4 Burn Assessment and Fluid Replacement Principles continued

Prolonged Field Care:

- The main target is a urine output of 30-50 ml/hr (0.5 ml/kg/hr) for adult and 0.5 ml/kg/hr to 1 ml/kg/hr for pediatric patients. Urine output needs to be monitored hourly and IV fluid rate should to be titrated to achieve target urine output.
- If UOP > 50 ml/hr, decrease IV fluid by 25% and reassess after 1 hr.
- If < 30 ml/kg, increase infusion volume by 25% for the next hour and reassess.
- The IV fluid should not exceed 1500 ml/hr x 2 hrs or a maximum of 250 ml/kg in 24 hrs to avoid over resuscitation/abdominal compartment syndrome.
- Antibiotics are not indicated for prophylaxis in the absence of open wounds. If after several days a cellulitis develops, contact the SMA.

Fluid Replacement Formula

Parkland Formula (for Paediatrics):

- 3 ml RL (or NS) x Weight in (kg) x TBSA with 2nd & 3rd degree burns in (%) = total ml given in the first 24 hrs.

  1/2 in 1st 8 hrs
  1/4 in 2nd 8 hrs
  1/4 in 3rd 8 hrs
  = Total in 24 hrs

- Example: Pt weighing 30 kg with 36% TBSA

  3 ml x 30 kg x 36 = 3240 ml/24 hrs

  1/2 in 1st 8 hrs = $\frac{3240 \text{ ml}}{2}$ = 1620 ml

  1/4 in 2nd 8 hrs = $\frac{3240 \text{ ml}}{4}$ = 810 ml

  1/4 in 3rd 8 hrs = $\frac{3240 \text{ ml}}{4}$ = 810 ml

- Pt would receive 1620 ml in the first 8 hours, 810 ml in the second 8 hours, and 810 ml in the last 8 hours for a total of 3240 ml in 24 hrs.
- If fluid was given to treat Hypovolemic/Hemorrhagic Shock, that amount needs to be subtracted from the total volume that will be administered in the 1st 8 hrs (e.g.: 1620 ml in the 1st 8 hrs - 500 ml given for Hypovolemic/Hemorrhagic Shock = the new volume for the 1st 8 hrs will be 1120 ml).
- Refer to Prolonged Field Care section above, for target urine output and titration.

USAISR Rule of Ten (For Adults):

- Initial IV/IO fluid rate is calculated as:
  a. %TBSA x 10 ml/hr (for adults weighing 40- 80kg);
  b. For every 10 kg ABOVE 80 kg, increase initial rate by 100 ml/hr;
  c. Refer to Prolonged Field Care section above, for target urine output and titration.